

NGUYEN NGUYEN NGOC ANH

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EDUCATION

- **University of Information Technology – Vietnam National University HCM** 2022 – Present
(Expected 2026)
Major: Software Engineering · GPA: 8.45/10.0

LANGUAGES

- **English** – TOEIC (Nov 2024): Reading & Listening 815

SKILLS

- **Programming:** TypeScript, JavaScript, Python
- **Frameworks:** ReactJS, NestJS, ExpressJS, FastAPI
- **Tools:** Git, VSCode, Postman, Figma, Draw.io
- **Soft skills:** English communication, team leadership, teamwork

PROJECTS

- **Thesis Management Platform**
Role: Co-Leader – Team size: 2
Stack: NestJS, ReactJS, MongoDB Atlas (Vector Search), LangChain, Socket.IO, Redis, Groq API
– Built an end-to-end platform digitizing the thesis lifecycle: topic registration, student–advisor matching, progress tracking, defense scheduling, and grading.
– Designed and implemented an agent-based RAG chatbot (LangChain + Gemini + ReAct) leveraging MongoDB Atlas Vector Search for semantic retrieval and AI-assisted Q&A.
– Developed a hybrid recommendation system (content-based + collaborative filtering) for student–advisor matching, with real-time notifications via Socket.IO and document storage using MinIO.
- **Restaurant Management Website**
Role: Leader – Team size: 4
Stack: ReactJS, Node.js/ExpressJS, MongoDB Atlas, Socket.IO, ZaloPay API, Figma, Draw.io
– Built a full-stack web application for restaurant operations: online ordering, table reservations, employee/inventory management, and revenue reporting.
– Implemented MVC architecture with real-time updates via Socket.IO, hybrid search for reservations, and secure ZaloPay payment integration.
– Defined use-cases for customer features (dish search, cart checkout, comments) and admin tools (order status, invoice creation, ingredient purchases).
- **Vietnamese News Search Engine**
Role: Solo Developer
Stack: Python, FastAPI, PostgreSQL, underthesea, BeautifulSoup4, psycopg2, uvicorn
– Crawled Vietnamnet.vn via XML sitemaps using concurrent **ThreadPoolExecutor** (10 workers); built a persistent **inverted index** (term → postings list) stored in PostgreSQL, mapping each term to its document frequency (df) and per-document term frequency (tf).
– Implemented **TF-IDF** ranking (log-normalized TF, corpus-level IDF) with **cosine similarity** scoring; pre-computed document length vectors and used batch upsert to eliminate N+1 database queries during indexing.
– Integrated a Vietnamese NLP preprocessing pipeline (underthesea word segmentation, diacritic normalization, stopwords filtering) and exposed a paginated search REST API via FastAPI.
- **AI-Chat: VSCode Extension (Cofire)**
Role: Co-Leader - Team size: 2
– Tool & Technology: TypeScript, HTML/CSS/JavaScript, Visual Studio Code API, ExpressJS, Figma, Postman, GitHub, AI Models (Gemini, etc.).
– Designed and implemented a VSCode extension integrating AI for code autocompletion, inline code generation, code explanation, and contextual chatting to enhance developer productivity.
– Built core features including AI model switching, context expansion (via files/folders), and seamless integration with VSCode’s editor, resulting in reduced coding time and improved code quality.